



BACHELOR OF COMPUTER SCIENCE (HONS)(SOFTWARE ENGINEERING)

FINAL EXAMINATION JANUARY 2019

Course : EC3307 (Object and Component Technology)

Time: 1:30pm-3:30pm
(2 hours)

Lecturer: Ms. R. Rajesvary

Date: 22 April 2019

Instructions:

Answer **ALL** questions in the Answer Booklet provided.
Each question carries 25 marks.

The maximum number of marks is 100.

This question paper consists of 5 printed pages
(excluding front cover).

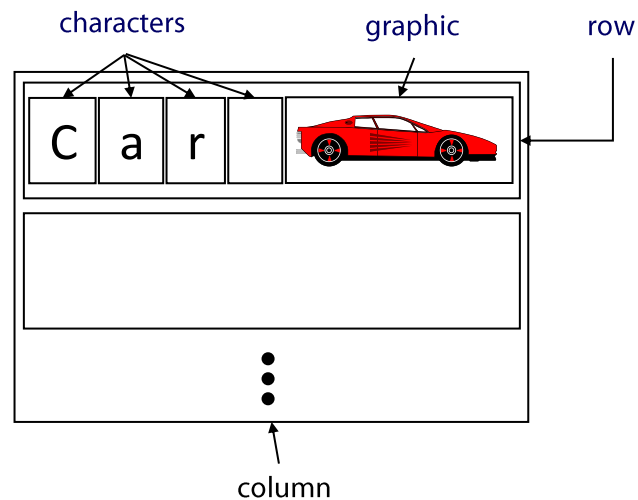
**Cheating is a very serious offence and those caught will be severely punished
DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO**

Answer all questions.

Question 1

[25 MARKS]

- a) A certain document editor structures each document so that the contents can be arranged in columns, with each column consisting of a number of rows. Each row can consist of a number of characters, spaces and graphic objects, as shown below:



All objects in a document are described as glyphs. The editor will call the following operations on the objects contained in the document:

```
void draw(Window)
//Draws this in window Window
```

```
boolean intersects(Point)
//Returns true if Point is within the bounding box of this object. Returns false otherwise.
```

```
void insert(Glyph, Pos)
//If this is a row or a column object, this method will insert Glyph at position Pos. Otherwise this method does nothing.
```

Note: the identifier **this** refers to the object on which the operation is called.

It is required that objects that form the internal representation of the document being edited, reflect the physical structure of the document as seen on the screen (e.g. columns and rows). It is also required that the class interface for rows and columns be the same as that for individual glyphs (e.g. characters).

- i) Identify the design pattern would you use to represent the contents of the document? Explain your answer.

(5 marks)

- ii) Based on the design pattern you used in part a(i) of this question, draw the class diagram for representing the glyphs. Include in your diagram the appropriate methods of each class.
(8 marks)
- b) Give an example situation when you might use a Singleton pattern and explain the major features of the code required. You do not need to write any actual code.
(5 marks)
- c) A goal of design patterns is to remove dependencies between classes. Briefly describe a general principle underlying all design patterns works to achieve this goal. Illustrate your answer with an example.
(7 marks)

Question 2

[25 MARKS]

- a) What is the difference between the specification of a method and its implementation?
(4 marks)
- b) A student sitting an exam gives the following subtly wrong definitions for pre- and post-conditions

“A precondition of a method is something that is always true whenever the method is entered.
A post-condition is something that is always true when the method is completed.”

Explain why both definitions are wrong.
(4 marks)
- c) The java.util.List interface has a get(int index) method which returns the element at the specified position in this list.
 - i) Write pre- and post-conditions in English for this method using the variables index and result, to indicate the value returned.
(4 marks)
 - ii) Explain any **ONE (1)** support that Java provides for checking pre- and post-conditions? Illustrate your answer by using suitable example.
(5 marks)
- d) Explain the meaning of the following JML keywords:
 - i) requires
 - ii) \old,
 - iii) \result; and
 - iv) pure.

(8 marks)

Question 3

[25 MARKS]

- a) Study the below code and answer the following questions.

```
import java.util.*;
public class WordGenerator {
    private List words = new ArrayList();
    private Random rand = new Random();

    public WordGenerator() {
        words.add(new String("SURPRISE"));
        words.add(new String("REVIEW"));
        words.add(new String("FROG"));
        words.add(new String("AZALEA"));
        words.add(new String("ARCHER JEFFREY "));
        words.add(new String("DISASTER"));
        words.add(new StringBuffer("VISIBLE"));
    }
    public String generateWord(){
        int size = words.size();
        String str1 = (String) words.get(rand.nextInt(size));
        String str2 = (String) words.get(rand.nextInt(size));
        String str3 = (String) words.get(rand.nextInt(size));
        return str1 + " " + str2 + " " + str3 + "!";
    }
}
```

- i) Explain what the WordGenerator class does. (2 marks)
- ii) Explain why there is an error in the code. (3 marks)
- iii) When will the error be discovered (assume that we ignore hints and warnings from the compiler and only take action when something fails to work)? (1 mark)
- iv) How would your answer to (iii) would be if we added a larger number of elements to the ArrayList words? (1 mark)
- v) How can we improve on our WordGenerator class by using generics? (3 marks)

b) Examine the following code:

```
import java.util.*;

public class ListPrinterDemo {
    public static void main(String[] args) {
        ListPrinter printer = new ListPrinter();
        List<String> myList = new LinkedList<String>();
        myList.add("Boundless and bare");
        myList.add("The lone and level sands");
        myList.add("Stretch far away");

        printer.printList(myList);
    }
}

import java.util.*;
public class ListPrinter {
    public void printList(List<String> list) {
        int size = list.size();
        for (int i=0; i<size; i++) {
            System.out.println(list.get(i));
        }
    }
}
```

- i) Rewrite the printList() method using an Iterator. (5 marks)
 - ii) Explain why using Iterator is a better idea than using the get(index) method. (4 marks)
- c)
- i) Explain by what a *cast* is in programming. (3 marks)
 - ii) Explain by using a suitable example under what conditions can a cast be used (3 marks)

Question 4

[25 MARKS]

- a) Multidimensional arrays in C# can be either jagged or rectangular.
 - i) Differentiate jagged and rectangular array. (6 marks)
 - ii) Discuss the relative merits of using jagged arrays over rectangular array. (6 marks)
- b) Explain an advantage of the following features of the .NET platform.
 - i) Common Type System (CTS) (4 marks)
 - ii) Microsoft Intermediate Language (MSIL) (4 marks)
- c) Explain why the limited reintroduction of the goto in C# is not considered harmful if used wisely. (5 marks)

-THE END-